Laboratorio di Elaborazione di Bioimmagini

Homework 3 – 19 December 2025

To upload the files in the proper folder in WeBeep (“cartella di consegna”) by **January 10th 2026 at 23:59**. Please post in the Forum discussion for doubts or comments.

I remind you that the solution is expected to be personal. If you discuss with a particular colleague in general term how to solve the problem is fine, but you need to declare it in the report.
In case of problems with upload, send an email to enrico.caiani@polimi.it

Exercise 1

Given the file *CMR_thorax.mat*, containing the variable vol representing a series of 104 images (uint16) of the thorax of a patient, write the code (and a brief report to explain and justify the choices made) that performs in the most automated way (i.e., minimizing the interaction with the user) the segmentation of the external contour of the thorax for each slice, and of the lungs in each slice in which they are visible, even partially.

Result of lung segmentation has to be saved in a 3D binary array called IcatLUNG, of the same dimensions of vol, where for each slice the pixels in the segmented lung are set to 1. In the same way, the segmented thorax has to be saved in a 3D binary array named IcatT.

At the end of the segmentation, the code should give as a result the total volume for left and right lung separately, computed as number of pixel, and visualize for each slice the contours of the thorax in yellow, and of the lungs (when present) in magenta, both overlapped to the original image.

For this exercise, you need to **provide the .m code and a short report in .pdf** with the results you obtained, and your final considerations.

Note:

The choice of the segmentation method, and its strategy (to operate in 2D or in 3D) is left to the student.

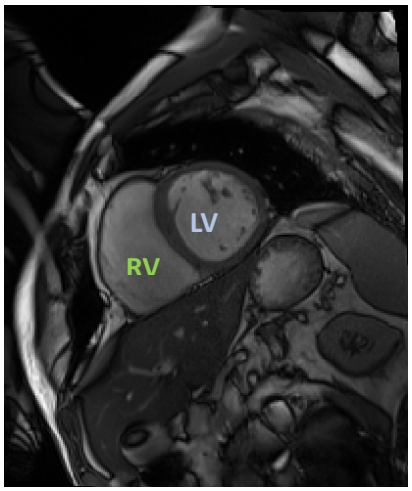
The use of proper morphological operators when needed is recommended. Also, use of morphological operators on the original graylevel images could be considered to better highlight structures to be segmented.

As a suggestion, it could be useful to start from the detection of the thorax, so to use it also to define internally a region of interest in which to segment the lungs.

An optimal detection of the lungs in all slices is not mandatory and not expected, but a good detection in a large number of them yes!

Exercise 2

Given the file *h3_loopMRI.mat*, containing the variable slice6, representing the series of 25 images (uint16) acquired during a cardiac beat for a short-axis section in a patient, write the code (and a brief report to explain and justify the choices made) that, in the most automated way (i.e., minimizing the user interaction), attempts to perform the segmentation of the endocardium of the left ventricle (LV) for each frame (please consider that the papillary muscles that you see inside the cavity should be considered as part of the cavity).



The coordinates of the LV contour for each frame have to be extracted in clockwise direction and saved.

At the end of the segmentation, the code should visualize the plot of the LV volume over time for the analyzed slice, computed in ml (use variables xres, yres e zres containing resolution in ml), and visualize for each frame the endocardial contours overlapped in green on the original image.

For this exercise, you need to **provide the .m code and a short report in .pdf** with the results you obtained, and your final considerations.

Prof. Enrico G Caiani